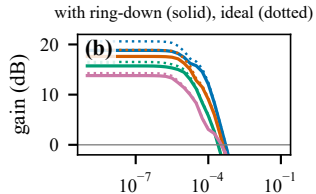
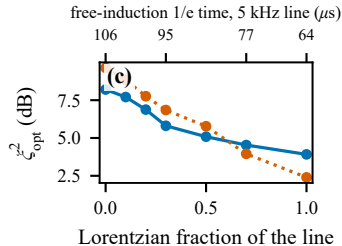


$g\sqrt{N}/\Delta$

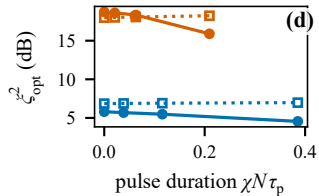
- exact, $N = 10$
- exact, $N = 20$
- $\propto (g\sqrt{N}/\Delta)^2$



$\kappa/2\pi = 3 \text{ kHz}$ $\kappa/2\pi = 30 \text{ kHz}$
 $\kappa/2\pi = 10 \text{ kHz}$ $\kappa/2\pi = 100 \text{ kHz}$



● echo twist ● free twisting



● loop-gap, echo
 □ loop-gap, free
 ● SC operating point, echo
 □ SC operating point, free